RAJALAKSHMI ENGINEERING COLLEGE

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H2

Started on	Saturday, 10 October 2026, 1:46 PM
State	Finished
Completed on	Saturday, 10 October 2026, 1:48 PM
Time taken	1 min 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int N1, N2;
        scanf("%d", &N1);

        int *a = malloc(N1 * sizeof(int));

        for (int i = 0; i < N1; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &N2);

        int *b = malloc(N2 * sizeof(int));

        for (int i = 0; i < N2; i++) {
            scanf("%d", &b[i]);
        }

        int i = 0, j = 0, first = 1;

        while (i < N1 && j < N2) {
            if (a[i] == b[j]) {
```

```
            if (!first)
                printf(" ");

            printf("%d", a[i]);
            first = 0;

            int value = a[i];

            while (i < N1 && a[i] == value)
                i++;

            while (j < N2 && b[j] == value)
                j++;
        } else if (a[i] < b[j]) {
            i++;
        } else {
            j++;
        }
    }

    printf("\n");

    free(a);
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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